Tyler Chisholm

Real-Time 3D Developer · UE5 Simulation

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UE5 C++ developer with a real-time rendering and VR background, graduating December 2026. I build interactive 3D systems end to end (physics, streaming worlds, VR input, tools) and measure performance with real instrumentation. U.S. citizen, eligible for a security clearance.

Education

M.S. Interactive Entertainment, Florida Interactive Entertainment Academy (FIEA), University of Central Florida. Dec 2026
12 games built across 4 semesters in UE5, Unity and VR.

B.B.A. Finance & Entrepreneurship, Stetson University, DeLand, FL May 2024

Skills

Real-Time	UE5 C++ and Blueprint, World Partition streaming, Chaos physics, OpenXR/VR, Live Link, GPU profiling, performance budgeting
Engines	Unreal Engine 5, Unity, Godot (engine fork), custom Vulkan renderer
Languages	C++, C#, Rust, HLSL, Python
Tools	Perforce, Git, Jira, Visual Studio, Rider, Nsight Graphics, FFmpeg

Experience

Design Lead — Everent (Studio Kismet, FIEA capstone)

UE5.7 · C++ · Blueprint · Chaos · Perforce Oct 2025 – Aug 2026

Momentum-based hoverboard game across an open world of floating sky islands in UE5.7 (Nanite, Virtual Shadow Maps, World Partition), coming to Steam.

- Designed and built speed-gated destruction in C++ on Chaos Geometry Collections: a predictive sweep measures impact speed into the wall, drops blockers a frame early to keep momentum, and fires strain, directional and radial velocity fields. It doubles as secret gating and skill expression.
- Built the Destructible Wall Browser editor tool: lists every destructible in the level, exposes its tuning and fires a test projectile at a set point and speed, so designers tune without entering Play mode.
- Hardened destructibles for World Partition and Data Layer streaming (delegate double-bind fix, overlap resolution at BeginPlay, streaming-safe destroy guards).
- Built a local playtest analytics pipeline (FFmpeg, faster-whisper, DeepFace, local Llama 3.1) that turns session recordings into transcripts, emotion timelines and coded feedback in the studio's Excel format.

Project Lead — Studio Microdose (FIEA Ventures)

Multiplayer co-op · publisher-pitch vertical slice Aug 2026 – Present

- Leading a 10-person team on a co-op game set in the Florida Everglades where players catch invasive animals by hand; hands-on in the build while running scope and leads (art, design, tech, community).

Selected Projects

Spatial Engine — path-traced VR engine

Rust · Vulkan RT · HLSL · OpenXR 2026 – Present

Personal engine, ~155k lines and 680+ commits. I own the architecture, direction and verification; implementation is AI-assisted.

- Runs the full path tracer in VR at 90 Hz on a Quest over Link on modest RTX hardware, across ~900 km of streamed terrain with volumetric clouds.
- VR climbing slice on the real Grand Canyon: terrain from USGS 3DEP lidar with 10 geologic formations as strata, tuned against camera-solved real photos.
- Physics-driven bodies: a Jolt active-ragdoll player with a contact-based support model for walking and climbing, and powered-ragdoll NPCs whose wound list (stab, stick, carve, fractures) drives bleeding and knockouts.

Star Fox Adventures on PS Vita — native port

C · GXM · PS Vita · released 2026

- Native PS Vita port built on the community SFA decomp: the decompiled game runs on the Vita's ARM CPU, and I wrote the graphics backend that translates the GameCube's GX API to GXM.

Earlier Experience

Quality Assurance Agent — Stormbreaker Studios, Orlando, FL

Apr – Jul 2020

Tested 8 iOS client apps; wrote 85 defect tickets with repro steps and fix recommendations.